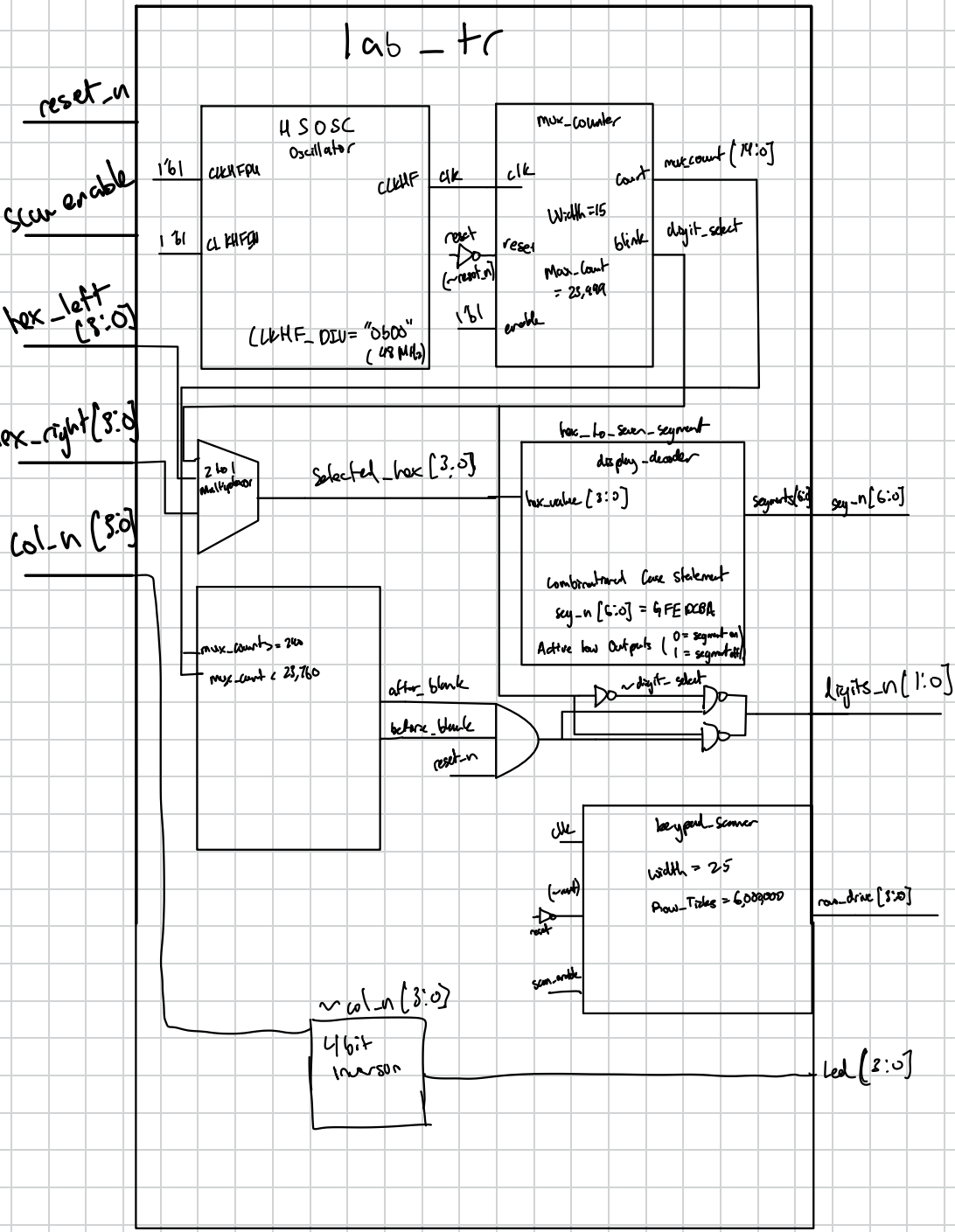

lab_tr
reset_n
scan_enable
hex_left [3:0]
hex_right [3:0]
col_n [3:0]
HSOSC
Oscillator
1'b1
CLKHFPU
CLKHF
1'b1
CLKHFEN
CLKHF_DIV = "0b00"
(48 MHz)
clk
mux_counter
clk
count
mux_count [14:0]
Width=15
reset
blink
digit_select
Max_Count
= 25,999
1'b1
enable
reset
(~reset_n)
2 to 1
multiplexer
selected_hex [3:0]
hex_to_seven_segment
display_decoder
hex_value [3:0]
segments[6:0]
seg_n[6:0]
Combinational Case Statement
seg_n [6:0] = 7'hFE0CBA
Active low Outputs (0 = segment on
1 = segment off)
mux_count >= 24a
mux_count < 25,760
after_blank
before_blank
reset_n
~digit_select
digits_n[1:0]
clk
keypad_scanner
width = 25
Row_Ticks = 6,000,000
(~rst)
reset
row_drive [3:0]
scan_enable
~col_n [3:0]
4bit
inverter
led [3:0]